

HARSHIT MOHINDRA

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EDUCATION

B.E. Electronics & Communication Engineering 2023 – 2027 (Expected)
University Institute of Engineering & Technology (UIET), Panjab University, Chandigarh

Overall CGPA: 7.44 | **Last Semester CGPA:** 7.9

Senior Secondary (12th) S.B. Senior Secondary School, Amritsar

Secondary (10th) D.A.V. Public School, Amritsar

INTERNSHIP EXPERIENCE

Embedded Systems Intern Summer 2024 (1 Month)

Design Innovation Centre (DIC), Panjab University × PGI Chandigarh

- Contributed to the development of a prosthetic hand as part of a biomedical embedded systems project.
- Designed and implemented a CNN-based onboard classification model to interpret sensor signals from the prosthetic hand.
- Achieved efficient real-time inference by deploying the model using Python and Embedded C on an ESP-32 microcontroller, enabling low-latency onboard processing without cloud dependency.

PROJECTS

Driver Drowsiness Detection & Vehicle Safety System (ongoing personal project)

- Built a computer-vision system that detects driver fatigue and drowsiness in real time using facial landmark analysis.
- Integrated with vehicle ECU logic: upon detecting drowsiness, the system triggers hazard/parking lights and initiates engine shut-off — functioning as a passive safety feature.
- Tech stack: Python, OpenCV, Arduino Uno

Indoor Navigation System using BLE Sensors (Certificate Obtained)

- Researched and prototyped an indoor positioning system leveraging Bluetooth Low Energy (BLE) beacons for navigation in GPS-denied environments.
- Achieved formal recognition for the work done; certificate obtained as proof of contribution and participation.

Prosthetic Hand with CNN Onboard Classification DIC × PGI Chandigarh (Certificate Obtained)

- Designed the signal-processing and ML pipeline for a prosthetic hand; implemented onboard CNN inference on ESP-32 for real-time gesture classification.

Car Parking Management System

- Designed and coded a parking slot management system using Arduino UNO and a 16×2 LCD display; handled circuit connections and firmware.

SAP Computer (Simple-As-Possible Computer)

- Implemented a basic CPU from scratch performing arithmetic operations in Assembly Language, reinforcing fundamentals of digital design and microarchitecture.

TECHNICAL SKILLS

Microcontrollers & Embedded: ESP-32, Arduino UNO, ATmega8/328, 8085, 8051 — programming in Embedded C & Python

Machine Learning: CNN model design & deployment, onboard inference optimisation

Languages & Tools: Python, Java, Verilog, Arduino IDE (Embedded C), HTML, CSS, JS (basics)

Coursework: Microprocessors & Microcontrollers, Digital Design, Data Structures & Algorithms

EXTRA-CURRICULAR & ACHIEVEMENTS

- Sub-Head of Filmmaking at Imagen (college film club); responsible for planning and executing video productions.
- Video editing and graphic designing for multiple college clubs and events.
- Keen interest in emerging technologies — regularly follows developments in embedded systems, ML, and hardware.

LANGUAGES

English (Fluent) | Hindi (Fluent) | Punjabi (Native)